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Psychiatric diagnoses in relation to severity of intellectual disability and challenging behaviors: a register study among older people

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ABSTRACT

Objective: To investigate the possible association between severity of intellectual disability (ID) and presence of challenging behavior, respectively, on diagnoses of psychiatric disorders among older people with ID.

Methods: People with a diagnosis of ID in inpatient or specialist outpatient care in 2002–2012 were identified ($n = 2147$; 611 with mild ID, 285 with moderate ID, 255 with severe or profound ID, and 996 with other/unspecified ID). Moreover, using impairment of behavior as a proxy for challenging behavior, 627 people with, and 1514 without such behavior were identified.

Results: Severe/profound ID was associated with lower odds of diagnoses of psychotic, affective, and anxiety disorders than was mild/moderate ID. People with moderate ID had higher odds than those with mild ID of having diagnoses of affective disorders. Diagnoses of psychotic, affective, and anxiety disorders, and dementia were more common among people with challenging behavior than among those without.

Conclusions: People with severe/profound ID had lower odds of receiving psychiatric diagnoses than those with mild and moderate ID. Whether this is a result of differences in prevalence of disorders or diagnostic difficulties is unknown. Further, challenging behaviors were associated with diagnoses of psychiatric disorders. However, the nature of this association remains unclear.

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Intellectual disability;
challenging behavior;
screening and diagnosis

Introduction

Psychiatric disorders are more common among people with intellectual disability (ID) than in the general population (Axmon, Björne, Nylander, & Ahlström, 2017; Cooper et al., 2007, 2015; Garaigordobil & Perez, 2007; Nettelbladt, Goth, Bogren, & Mattisson, 2009). In spite of the well-known difficulties in diagnosing psychiatric disorders in people with ID, especially those with moderate to profound ID, attempts have been made to investigate a possible association between prevalence of psychiatric disorders and severity of ID. Most published data point to a negative association between severity of ID and prevalence of psychiatric disorders (Holden & Gitlesen, 2004; Hurley, Folstein, & Lam, 2003; Myrbakk & von Tetzchner, 2008; Nettelbladt et al., 2009; Tsiouris, Kim, Brown, & Cohen, 2011). Although there are inconsistencies across studies, some have found this pattern to be repeated for affective (Holden & Gitlesen, 2004; Hurley et al., 2003; Myrbakk & von Tetzchner, 2008; Tsiouris et al., 2011), anxiety (Holden & Gitlesen, 2004; Hurley et al., 2003; Tsiouris et al., 2011), psychotic (Holden & Gitlesen, 2004; Myrbakk & von Tetzchner, 2008; Tsiouris et al., 2011), and personality disorders (Tsiouris et al., 2011).

The term ‘challenging behavior’ was introduced by the Association for Severe Handicaps with the intention of transferring the demands for change from the individuals to the organization surrounding them. Challenging behavior may present itself in various ways, e.g. through aggression, self-injury, or inappropriate social conduct. However, challenging behavior in itself is not a disorder or diagnosis. Challenging

behaviors are common among people with ID (Bowring, Totsika, Hastings, Toogood, & Griffith, 2017; Cooper, Smiley, Allan, et al., 2009; Cooper, Smiley, Jackson, et al., 2009; Schützwohl et al., 2016; Sheehan et al., 2015), and have been suggested to be associated with psychiatric disorders (Deb, Thomas, & Bright, 2001; Felce, Kerr, & Hastings, 2009; Myrbakk & von Tetzchner, 2008). Nevertheless, studies regarding possible associations between challenging behaviors and specific psychiatric disorders are scarce. As challenging behavior can be of serious consequence for people with ID, analysis of their relation to psychiatric disorders is of importance.

The prevalence of challenging behaviors has been found to be associated with severity of ID (Crocker, Mercier, Allaire, & Roy, 2007; Deb et al., 2001; Holden & Gitlesen, 2006; Tsiouris et al., 2011). Hence, in the study of either challenging behaviors or severity of ID, it is important to take the other factor into account.

The life expectancy among people with ID, and consequently the number of people with ID who reach older age, is increasing (Coppus, 2013; Dieckmann, Giovis, & Offergeld, 2015; Haveman, 2004; Patja, Iivanainen, Vesala, Oksanen, & Ruoppila, 2000). This group of elderly people can be expected to be especially mentally frail as it is common with cognitive deterioration, dementia, and polypharmacy, as well as living with the experience of multiple adverse life events (Hulbert-Williams & Hastings, 2008), and some psychiatric diagnoses have been found to be more common among older people with ID than their younger peers (Cooper, 1997). Yet, to the best of our knowledge, no study has focused on older people

with ID when assessing the possible association between on one hand severity of ID or challenging behaviors and on the other psychiatric disorders and diagnoses. As life expectancy is negatively associated with severity of ID (Coppus, 2013; McCarron, Carroll, Kelly, & McCallion, 2015), people with severe or profound ID will be underrepresented in older age groups. The association between challenging behavior and age among people with ID is unclear, with one study claiming it to peak among those 70+ years (Lundqvist, 2013), and another finding it more common in younger people (Holden & Gitlesen, 2006). The complex situation involving severity of ID, challenging behavior, older age, and coexisting psychiatric disorders needs to be elucidated in order for policy-makers, health organizations, and service providers to be able to provide support and service for aging people with ID.

The aim of the present study was to investigate psychiatric diagnoses among older people with ID, in relation to severity of ID and presence of challenging behavior. A further aim was to assess a potential combined effect (interaction) of severity of ID and challenging behavior on psychiatric diagnoses.

Methods

The data collection in the present study was performed in three steps (Figure 1). First, the LSS register was used to establish a cohort of people with ID, aged 55+ years and alive at the end of 2012. The LSS register contains information on all support and services given to people with ID and autism spectrum disorder (ASD) according to the Swedish Act Concerning Support and Service for People with Certain Functional Impairments (the LSS act)(SFS 1993:387, 1993). To receive such support, the individual must apply for it and an officer at the municipality must assess the individual as being entitled to it. Entitlement is considered, among other things, based on diagnosis of either ID or ASD. However, the diagnosis is not recorded in the register. Thus, it is not possible to separate those with ID only from those with ASD only by using the LSS register.

Second, people with at least one recorded diagnosis of ID (F7 in the International Statistical Classification of Diseases and Related Health Problems, 10th revision; ICD-10) during the period 2002–2012 were identified through the Swedish National Patient Register (NPR; Figure 1). This register contains

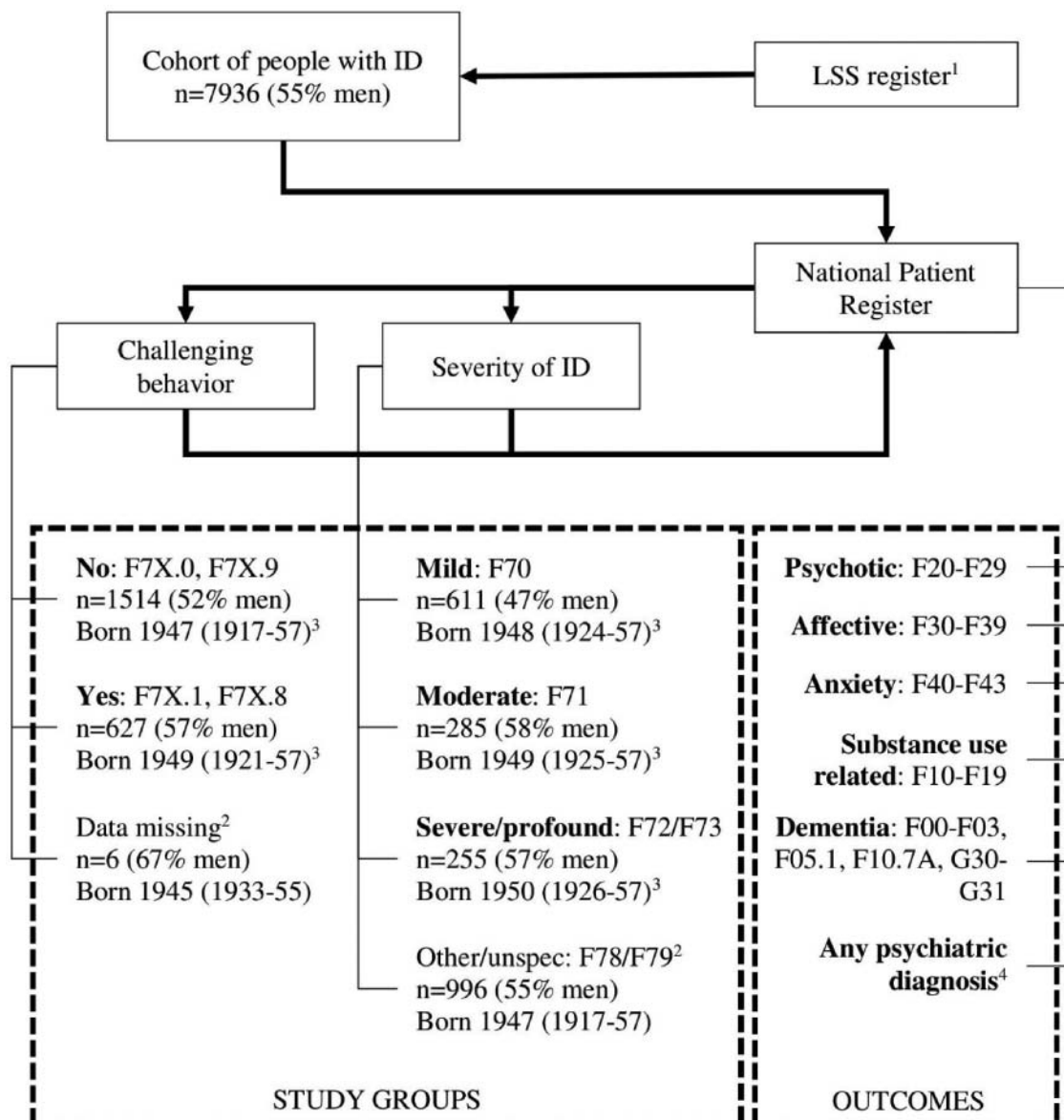


Figure 1. Flow chart of the data collection process and characteristics of the different sub-groups in the present study.

information about all inpatient and outpatient specialist care in Sweden. However, it does not contain information on visits to primary care (including family physicians and general practitioners). All information in the register was used, i.e. we did not restrict the data collection to any particular type of specialist care. People with an ID diagnosis were categorized in two ways: by severity of ID (mild, moderate, and severe/profound) and by presence of challenging behavior. Each person was classified according to the most severe degree (combining severe and profound ID) of ID recorded during the study period. Thus, a person with one F70 diagnosis and one F72 diagnosis was classified as F72. Thus, 59 people who were classified as having moderate ID at one point also had a diagnosis of mild ID, 24 people classified as having severe/profound ID were also recorded as having moderate ID, and a further 8 as having mild ID. People with only F78/F79 diagnoses were included in the analyses of challenging behavior, but not in analyses regarding severity of ID.

Although there is no code for challenging behavior in ICD-10, each ID diagnosis may be classified as 'With the statement of no, or minimal, impairment of behavior' (F7X.0), 'Significant impairment of behavior requiring attention or treatment' (F7X.1), 'Other impairments of behavior' (F7X.8), or 'Without mention of impairment of behavior' (F7X.9), where impairment of behavior suggests repetitive self-injury, pica, hyperkinesia, wandering and absconding, or aggression towards other (WHO, 1996). We used this classification as proxy for challenging behavior, and will use the term 'behavior impairment' when referring to the results obtained in the present study.

The third step was to collect information on psychiatric diagnoses from the NPR for the people with a diagnosis of ID (Figure 1). Psychiatric diagnoses were categorized as psychotic, affective, anxiety, and alcohol/substance use related disorders, and dementia. All such diagnoses were included, regardless if they were primary or secondary, or from psychiatric or somatic clinics.

Statistics

Each person was categorized as having none or at least one diagnosis in each diagnostic category (psychotic, affective, anxiety, and alcohol/substance use related disorders, and dementia) during the study period. Moreover, each person was classified as having none or at least one psychiatric diagnosis including the diagnoses described above as well as 'any psychiatric disorder' (Figure 1).

Analyses were performed using logistic regression, whereby odds ratios (ORs) with 95% confidence intervals (CIs)

were estimated. All analyses were adjusted for gender and year of birth. The possible statistical interaction between severity of ID and challenging behavior was evaluated by adding the cross-product between these two variables to the logistic regression model. Wherever such interaction was present, stratified results are presented.

As there is a known overlap between ASD and ID, and ASD is associated with several psychiatric diagnoses, we performed additional analyses in which we evaluated ASD as a potential confounder and effect modifier.

In order to assess possible bias from including people based on recorded diagnoses, we performed sensitivity analyses comparing those with at least one F7-diagnosis to those without any such diagnosis. In these analyses, we included only those with at least one visit recorded in the NPR, as those who never visited inpatient or outpatient specialist care could not get an F7-diagnosis. Furthermore, in order to assess potential misclassification due to poorly made diagnoses, we performed sensitivity analyses excluding those with records of more than one severity of ID.

ORs were only estimated when each compared group contained at least five individuals. A two-tailed *p* value of 0.05 was considered statistically significant. All analyses were performed in IBM SPSS Statistics 23.

Ethical considerations

Approval was obtained from the Regional Ethical Review Board in Lund (2013/15). The National Board of Health and Welfare as well as Statistics Sweden performed separate privacy reviews in 2014 before providing access to the data. All analyses were performed using anonymized data-sets.

Results

Out of the 7936 people in the original cohort, at least one diagnosis of ID was found for 2147 (27%; Figure 1). The majority of these (46%) had diagnosis of other or unspecified ID, followed by mild ID (28%), moderate ID (13%), and severe/profound ID (12%).

All psychiatric diagnoses, except dementia, were less common among those with severe/profound ID than among those with mild or moderate ID, as was having at least one psychiatric diagnosis during the study period (Table 1). No such clear pattern was found when comparing those with moderate to those with mild ID, where the only statistically significant effect was an increased odds of affective disorders among people with moderate ID. Concomitant ASD diagnosis was not uncommon, especially in those with severe/profound

Table 1. Number of people with at least one diagnosis in different psychiatric diagnostic categories during 2002–2012 among people with mild (F70 in ICD-10, *n* = 611, reference), moderate (F71, *n* = 285), and severe/profound (F72 and F73, *n* = 255) intellectual disability. Statistically significant comparisons are marked in bold.

	Moderate ID			Severe/profound ID		
	Mild ID <i>n</i> (%)	<i>n</i> (%)	Moderate vs. mild OR (95% CI) ^a	<i>n</i> (%)	Severe vs. mild OR (95% CI) ^a	Severe vs. moderate OR (95% CI) ^a
Psychotic disorders	153 (25)	66 (23)	0.91 (0.65–1.27)	19 (7)	0.24 (0.15–0.40)	0.27 (0.16–0.46)
Affective disorders	153 (25)	90 (32)	1.47 (1.07–2.01)	29 (11)	0.40 (0.26–0.61)	0.28 (0.17–0.44)
Anxiety disorders	124 (20)	52 (18)	0.87 (0.60–1.25)	21 (8)	0.34 (0.21–0.55)	0.39 (0.23–0.68)
Personality disorders	20 (3)	4 (1)	NC	1 (0)	NC	NC
Alcohol/substance use related dis.	36 (6)	9 (3)	0.50 (0.24–1.06)	2 (1)	NC	NC
Dementia	32 (5)	8 (3)	0.60 (0.27–1.33)	6 (2)	0.51 (0.21–1.24)	0.85 (0.29–2.47)
Any psychiatric diagnosis ^b	407 (67)	188 (66)	0.96 (0.71–1.30)	137 (54)	0.56 (0.41–0.75)	0.59 (0.41–0.83)

Note: NC = not calculated (less than five observations per group).

^aOdds ratio (OR) with 95% confidence intervals (CIs) estimated by logistic regression, adjusted for year of birth and gender.

^bAll diagnoses listed above, as well as hyperkinetic disorders (F90), pervasive developmental disorders (F84), and 'other' psychiatric disorders (F05–F09, F44–F48, F50–F53, F63, F65, F68, F69, F80, F81, F83, F89, F91–F99, and R440).

Table 2. Number of people with at least one diagnosis in different psychiatric diagnostic categories during 2002–2012 among people with intellectual disability with (ICD-10 diagnosis F7X.1 or F7X.8, $n = 627$) or without (F7X.0 or F7X.9, $n = 1514$) challenging behavior. Statistically significant comparisons are marked in bold.

	Without	With	
	n (%)	n (%)	OR (95% CI) ^a
Psychotic disorders	197 (13)	120 (19)	1.56 (1.21–2.01)
Affective disorders	211 (14)	162 (26)	2.16 (1.71–2.72)
Anxiety disorders	129 (9)	99 (16)	1.69 (1.28–2.23)
Personality disorders	19 (1)	10 (2)	1.22 (0.56–2.64)
Alcohol/substance use related disorders	46 (3)	19 (3)	0.92 (0.53–1.59)
Dementia	53 (4)	40 (6)	2.21 (1.43–3.41)
Any psychiatric diagnosis ^b	700 (46)	448 (71)	2.83 (2.31–3.46)

^aOdds ratio (OR) with 95% confidence intervals (CIs) estimated by logistic regression, adjusted for year of birth and gender.

^bAll diagnoses listed above, as well as hyperkinetic disorders (F90), pervasive developmental disorders (F84), and 'other' psychiatric disorders (F05–F09, F44–F48, F50–F53, F63, F65, F68, F69, F80, F81, F83, F89, F91–F99, and R440).

ID (26%), and also among those with moderate (12%) or mild ID (6%) as well as those with other or unspecified ID (12%). Including ASD in the model changed the effect estimates only marginally, and did not affect the statistical significance of any of the estimates. Moreover, there were no statistically significant interactions between ASD and severity of ID for any of the outcomes investigated.

Challenging behaviors were more common among those with severe/profound (49%) and moderate (49%) ID than among those with mild (25%) or other/unspecified (21%) ID. People with challenging behaviors were almost three times more likely than those without to have at least one psychiatric diagnosis recorded during the study period (Table 2). Excess odds were also found for diagnoses of psychotic, affective, and anxiety disorders, as well as for dementia diagnosis. Among those with challenging behavior, 21% had also a diagnosis of ASD. Such a diagnosis was present among only 9% of those without challenging behavior. Including ASD in the model changed the effect estimates only marginally, and did not affect the statistical significance of any of the estimates. There was a statistically significant interaction between ASD and challenging behavior with respect to anxiety disorders such that the increased risk for having a diagnosis of anxiety disorders associated with challenging behavior was statistically significant among those without ASD (OR 1.66, 95% CI 1.22–2.26) but not among those with ASD (OR 1.85, 0.88–3.86). No other interaction was found between ASD and challenging behavior.

There was a statistically significant interaction between severity of ID and challenging behavior with respect to diagnoses of psychotic disorders ($p = 0.009$), but not for any of the other diagnoses investigated (data not shown). Among those with mild ID, there was an increased odds of being diagnosed with psychotic disorders associated with challenging behavior (OR 2.52, 95% CI 1.69–3.75). This was not found among those with moderate (OR 0.89, 0.51–1.55) or severe/profound ID (OR 1.58, 0.60–4.15).

There were 4371 people who had at least one record in the NPR during the study period, but never a diagnosis of ID. The sex distribution among these was similar among those with (53% men) and those without ID diagnosis (54% men). After adjusting for sex and year of birth, those with at least one ID diagnosis were more likely than those without to have received at least one diagnosis of psychotic disorders (OR 6.83, 95% CI 5.43–8.52), affective disorders (OR 4.29, 3.58–5.14), anxiety disorders (OR 2.85, 2.33–3.48), personality disorders (OR 1.82, 1.10–3.03), and dementia (OR 1.58, 1.20–2.08), but not alcohol/substance use related disorders (OR 1.08, 0.79–1.47).

Out of the 1151 people classified as having mild, moderate, or severe/profound ID, 611 had only diagnoses of mild ID, 226 only of moderate ID, and 223 only of severe/profound ID. When restricting the analyses to these, the increased OR for affective disorders for moderate vs. mild ID was attenuated (OR 1.09, 95% 0.76–1.55). For anxiety disorders, the OR for moderate vs. mild ID was decreased (OR 0.55, 0.35–0.86), but the OR for severe/profound vs. moderate ID was increased (OR 0.58, 0.31–1.09). Moreover, the decreased OR for any psychiatric diagnoses for severe/profound vs. moderate ID increased (OR 0.72, 0.49–1.05). In the remaining analyses, the statistical significance was not affected.

Discussion

We found lower odds of psychiatric diagnoses among people with severe ID compared with those with mild or moderate ID. People with moderate ID had lower odds of diagnoses of affective disorders compared with those with mild ID, but no other differences were found between these two groups. Challenging behavior was associated with higher odds of diagnosis of a range of psychiatric disorders. The majority of these associations were independent of severity of ID.

Severity of ID and presence of challenging behavior was determined using F7-diagnoses in the NPR during 2002–2012. Such diagnoses were available for only 27% of the total ID cohort. If these 27% are not representative for the entire cohort, we run the risk of introducing a selection bias. In the NPR, 1 primary and up to 21 secondary diagnoses are listed for each record. The primary diagnosis is supposed to reflect the cause of the care episode, whereas the secondary diagnoses are to give information about conditions that have been important for the diagnosing or treatment (SOSFS 2013:30, 2013). Conditions without relevance for the care episode should not be recorded. Thus, it is not given that a person with ID would have the ID recorded as a secondary diagnosis, should the physician not consider the ID relevant for the treatment of the primary diagnosis. Hence, that we failed to find an ID diagnosis for such a large part of the study cohort is not surprising. Moreover, given the age of the study group, it is unlikely that they received their first ID diagnosis during the study period. However, those with at least one ID diagnosis had higher odds of almost all the psychiatric diagnoses, suggesting that a bias may indeed be at play regarding who gets an F7-diagnosis recorded in inpatient or outpatient specialist care. Nevertheless, and more importantly, the distribution over different levels of severity of ID is similar to that found in previous studies in the Swedish population (Hällgren, Nygard, & Kottorp, 2014; Nettelbladt et al., 2009), possibly with a slight

shift towards moderate ID. Thus, even though the studied group may be less healthy than the population of older people with ID in Sweden, they seem to constitute a representative sample of people with ID with respect to severity of ID. Thus, there is little reason to believe that there is any bias regarding the comparisons within the studied group.

All people included in the present study at one point had support and service according to the LSS act. To be eligible for this support, the diagnosis of ID or ASD has to have been present since childhood. Thus, no one in the cohort should have had an ID evaluation during the study period. It is more likely that the ID diagnoses registered during the study period were considered relevant to either the cause of the health care visit or the treatment. Therefore, it is not surprising that some of the people in the cohort had more than one diagnosis of ID in the NPR during the study period. For these people, we used the lowest level of ID recorded to classify the person with respect to severity of ID. We believe this to be the best alternative as, based on clinical experience of one of the authors (L. Nylander), it is uncommon for people to be diagnosed with a lower level of ID than they actually have.

The quality regarding psychiatric care registrations in the NPR has improved during the study period in that specialist outpatient visits without a primary diagnosis have become less common (75% in 2002 vs. just above 10% in 2012; National Board of Health and Welfare, 2014). This impacts both the classification of severity of ID and challenging behavior, and the availability of outcome data (recorded psychiatric diagnoses). The lack of complete data, especially during the earlier years of the study, is most likely a contributor to the failure to classify people with respect to severity of ID and challenging behavior. Had more comprehensive data been available, we would presumably have been able to perform the analyses on larger groups, i.e. with higher statistical power.

We have chosen to use recognized behavior impairments (F7x.1 and F7x.8 in ICD-10) as a proxy for challenging behavior. If behavior impairment is always equal to challenging behavior is not clear from the coding, nor do we know if all patients with challenging behaviors have been included when using this proxy. However, clinical experience permits us to assume that challenging behaviors in ID are under-reported in health care registers rather than over-reported. The reason for this may be that challenging behaviors are assumed to be an intrinsic part of the ID. If this is so, it may also follow that if challenging behavior has been noted, it is likely to have been of a severity that brought the patient to a medical assessment. Moreover, it should be emphasized that people with challenging behaviors were not compared with all those without this coding, but to those with the coding of not having behavior impairments, i.e. when the diagnosing professional actively determined that the person did not have any behavior impairment.

In 1994, Borthwick-Duffy (1994) published a review of studies performed so far regarding severity of ID and psychiatric co-morbidity. Some of the results included pointed towards a positive association between severity of ID and prevalence of psychiatric disorders, whereas others suggested the opposite association, and a third group presented no association at all. Differences were suggested to arise from variations in methods of assessment, referral practices, or representativeness of samples. The conclusion of the author was that the results were conflicting, and that the relationship between

psychiatric co-morbidity and severity of ID had yet to be established. Since then, slightly conflicting data have been published regarding prevalence of psychiatric disorders and severity of ID (Cooper & Bailey, 2001; Gustafsson & Sonnander, 2004; Holden & Gitlesen, 2004; Hurley et al., 2003; Myrbakk & von Tetzchner, 2008; Nettelbladt et al., 2009; Strydom, Hassiotis, King, & Livingston, 2009; Tsiouris et al., 2011). However, existing literature provides evidence of a negative association between severity of ID and prevalence of psychiatric diagnoses overall, as well as for psychotic and personality disorders, and depression. For anxiety disorders, a decreased occurrence in the moderate to profound ID span would fit previously published data.

Although we found lower odds of having any psychiatric diagnosis, as well as any diagnosis of psychotic or affective disorder (in which depression is included), among those with severe/profound ID compared with mild ID, our results for moderate ID is in conflict with some of the previous studies (Holden & Gitlesen, 2004; Myrbakk & von Tetzchner, 2008; Nettelbladt et al., 2009). The discrepancy is most likely a result of differences in, e.g. age, definition of psychiatric diagnoses (e.g. ICD-10 vs. DSM-IV [Diagnostic and statistical manual of mental disorders, 4th edition]), and health care systems. Nevertheless, with respect to diagnoses of anxiety disorders, our results support previous data indicating similar occurrence among those with mild and moderate ID, and a decrease with more severe ID (Holden & Gitlesen, 2004). To the best of our knowledge, ours is the first study to present data regarding severity of ID and occurrence of diagnoses of dementia or alcohol/substance use related disorders.

In his 2008 review, Allen (2008) found that most studies trying to link challenging behavior to psychiatric co-morbidity among people with ID either found a positive association or no association at all. However, studies investigating specific psychiatric diagnoses were inconsistent. This caused him to conclude that research regarding challenging behavior and co-occurring mental ill-health had produced mixed findings. A few years later, Pruijssers, van Meijel, Maaskant, Nijssen, & van Achterberg (2014) reviewed the relationship between challenging behavior and anxiety among people with ID, and drew the conclusion that although such a relationship was present, its nature was unclear.

Studies performed since the review by Allen (2008) support the idea of a positive association between challenging behavior and psychiatric disorders (Felce et al., 2009; Grey, Pollard, McClean, MacAuley, & Hastings, 2010; Lundqvist, 2013; Tsiouris et al., 2011). The results seem to be fairly consistent when assessing different types of challenging behaviors, such as aggression and self-injurious behavior. With respect to the possible association between challenging behavior and specific psychiatric diagnoses, published data suggest that there may be a positive relationship between challenging behavior and depression, anxiety, and psychosis. This is well in line with the results from the present study, where increased occurrence of any psychiatric diagnosis as well as psychotic, affective, and anxiety disorder diagnoses were found among people with challenging behavior. In addition, we found an association between challenging behavior and diagnosis of dementia. To the best of our knowledge, this has not been studied previously.

The interaction found between severity of ID and challenging behavior may in part be due to a bias, in that behaviors indicating symptoms of psychosis are more readily identified

in persons with mild ID. Diagnostic overshadowing could to some extent explain that challenging behaviors are not identified as symptoms of psychosis in persons with severe ID. Rather, the behaviors could be attributed to the person's diagnosis of ID. On the other hand, there might be a risk that persons with mild ID are 'over-diagnosed' with psychosis, in that their behaviors are too commonly identified as symptoms of psychosis rather than a reaction of poor quality in the care environment.

It is important to note that even though the data in the present study were collected for a period of several years, they only represent a short time window in the lives of the people studied. As we cannot tell what happened before the data collection period started, we are not able to say if the challenging behavior diagnoses preceded the psychiatric disorder diagnoses. Thus, the present data do not allow for conclusions regarding causality, i.e. if challenging behavior increases the occurrence of psychiatric diagnoses or psychiatric diagnoses lead to increased occurrence of challenging behavior. Indeed, our data are consistent also with a third option, namely that challenging behavior and psychiatric disorders share a common cause.

Conclusion

Similar to what has previously been found in younger populations, older people with severe/profound ID had lower odds of receiving psychiatric diagnoses than those with mild/moderate ID. If this is a result of differences in prevalence of disorders, or diagnostic difficulties is unknown. Challenging behaviors were associated with diagnoses of psychiatric disorders. However, the nature of this association remains unclear. Further studies are needed to elucidate the overlap and potential interaction between challenging behaviors and severity of ID in order to provide optimal treatment for psychiatric disorders among people with ID.

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Disclosure statement

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